```
clc;
close all;
clear all;
% This Program has been Developed by Mohammed Lamine MOUSSAOUI
% VDA 6.3 / SAE J1211
% A Five Order Polynomial is used in the Nonlinear Regression Process
format short;
No=71 ;
Ns(1)=No ;
R(1)=Ns(1)/No ;
f(1)=0 ;
Lambda(1)=0 ;
x=[ 120 260 420 650 1100 1900 2900 3800 4700 5400 6100 7000 ] ;
n=length(x)
Nsdx=[ 3 6 8 4 2 2 2 5 12 14 9 4 ] ;
for i=2:n
    Deltax(i)=x(i)-x(i-1);
end
for i=2:n
    Ns(i)= Ns(i-1)-Nsdx(i-1) ;
    R(i)=Ns(i)/No ;
    F(i)=1-R(i);
    Nodx=No*Deltax(i);
    f(i)=Nsdx(i-1)/Nodx ;
    Lambda(i)=f(i)/R(i) ;
    xh(i-1)=x(i) ;
    fh(i-1)=f(i) ;
    Lambdah(i-1)=Lambda(i);
end
x
Ns
R
f
Lambda
plot(x,R)
title('Reliabilities  R(x)')
xlabel('x')
ylabel('R')
pause
plot(x,F)
title('Cumulative Distribution Function F(x) : Failure Probabilities')
xlabel('x')
ylabel('F')
pause
plot(xh,fh,'+',xh,fh)
title('Density function  f(x)')
xlabel('x')
ylabel('f(x)')
pause
plot(x,Lambda)
title('Failure Rate Lambda(x)')
xlabel('x')
ylabel('Lamda')
pause
np=n-1;
for i=1:np
    xi1=xh(i)  ;
    xi2=xi1*xi1 ;
    xi3=xi1*xi2 ;
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```
    xi4=xi2*xi2 ;
    xi5=xi2*xi3 ;

    Phi1(i)=1 ;
    Phi2(i)=xi1 ;
    Phi3(i)=xi2 ;
    Phi4(i)=xi3 ;
    Phi5(i)=xi4 ;
    Phi6(i)=xi5 ;
end
Phit=[ Phi1 ; Phi2 ; Phi3 ; Phi4 ; Phi5 ; Phi6 ] ;
Phi=Phit' ;
A=Phit*Phi;
Y=fh ;
B=Phit*Y' ;
r=A\B ;
a1=r(2) ;
a2=r(3) ;
a3=r(4) ;
a4=r(5) ;
a5=r(6) ;
for i=1:np
    xi1=xh(i)  ;
    xi2=xi1*xi1 ;
    xi3=xi1*xi2 ;
    xi4=xi2*xi2 ;
    xi5=xi2*xi3 ;
    fregression(i)= r(1)+ a1*xi1 + a2*xi2 + a3*xi3+ a4*xi4+ a5*xi5;
    Epsilon(i)=fh(i)-fregression(i) ;
end
n1=length(x)
n2=length(fregression)
Epsilon
plot(xh,fregression) ;
xlabel('x')
ylabel('Density Function fs(x)')
title('Regression fs(x)')
pause
nx=10*np ;
xdeb=x(1);
xfin=x(n);
pas=(xfin-xdeb)/(nx-1) ;
x0=xdeb-pas ;
for i=1:nx
    xi1=x0+pas  ;
    xi2=xi1*xi1 ;
    xi3=xi1*xi2 ;
    xi4=xi2*xi2 ;
    xi5=xi2*xi3 ;
    smoothregression(i)= r(1)+ a1*xi1 + a2*xi2 + a3*xi3+ a4*xi4+ a5*xi5;
    xs(i)=xi1 ;
    x0=xi1 ;
end
plot(xs,smoothregression) ;
xlabel('x')
ylabel('Density Function fsmooth(x)')
title('VDA 6.3 / SAE J1211')
legend('Smooth Regression fs(x)')
x1=x(1);
```

```
xn=x(n);
display(' Mean Time Between Failures MTBF = integral{x*f(x)*dx}')
Theta_m=(r(1)/2)*(xn^2-x1^2)+(a1/3)*(xn^3-x1^3)+(a2/4)*(xn^4-x1^4)+(a3/5)*(xn^5-
x1^5)+(a4/6)*(xn^6-x1^6)+(a5/7)*(xn^7-x1^7)
```